AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 20 · STUDENT LESSON

Image tensors and channels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why does a representation with two components not necessarily preserve two original measurements?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Read a pixel using declared row column and channel axes. Try the recall before opening notes.

Your learning target

I can read a pixel using declared row column and **channel** axes.

I can convert **channel** layout without changing pixel meaning.

I can distinguish fixed **value scaling** from information losing transformations.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Read a pixel using declared row column and **channel** axes

Convert **channel** layout without changing pixel meaning

Distinguish fixed **value scaling** from information losing transformations

Retrieve what you already know

> Why does a representation with two components not necessarily preserve two original measurements?

My starting explanation

This week's end goal

By the end, I can explain image **tensor**s and **channel**s using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Image tensors and channels case. Separate observations, given facts and assumptions before choosing an explanation.

A small fictional RGB image has height 2 pixels, width 2 pixels and three **channel**s ordered red, green, blue. Store it as image[row][column][**channel**], abbreviated HWC.

Use zero-based indices: the top row is 0, the left column is 0, and red, green and blue are **channel**s 0, 1 and 2. Each stored component is an integer from 0 to 255.

Position	RGB triplet	Meaning
Row 0 column 0	[255, 0, 0]	Red
Row 0 column 1	[0, 255, 0]	Green
Row 1 column 0	[0, 0, 255]	Blue
Row 1 column 1	[255, 255, 255]	White

The shape is 2 by 2 by 3. There are four pixels and twelve **channel** values. The blue component of the bottom-left pixel is image[1][0][2] = 255.

A single **channel** is a plane of values across positions; it is not a separate photograph or a label.

Shape needs a schema. A grayscale image might be stored as H by W or H by W by 1. An RGB image uses three **channel**s here; other formats can include additional **channel**s.

Declare the format instead of assuming every three-dimensional array means the same thing.

List the given facts and one fact that is not supplied.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Image tensors and channels. Connect each labelled input, deciding step and result.

A **channel**-first layout uses image[**channel**][row][column], abbreviated CHW. Our image then has shape 3 by 2 by 2. A correct conversion preserves the rule chw[c][r][k] = hwc[r][k][c].

The code assumes a nonempty rectangular image with exactly three values per pixel.

```
hwc = [
    [[255, 0, 0], [0, 255, 0]],
    [[0, 0, 255], [255, 255, 255]],
]
height, width = len(hwc), len(hwc[0])
chw = [[[hwc[r][k][c] for k in range(width)]
        for r in range(height)] for c in range(3)]
for channel in chw:
    print(channel)
assert chw[2][1][0] == hwc[1][0][2] == 255
restored = [[[chw[c][r][k] for c in range(3)]
             for k in range(width)] for r in range(height)]
assert restored == hwc
```

Reading nested values in order and merely regrouping them into a new shape does not perform this axis conversion. The first four flattened HWC values are [255, 0, 0, 0].

A naive regrouping makes these the first **channel** plane, but the correct red plane is [[255, 0], [0, 255]]. Shape agreement alone cannot detect this mistake.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Image tensors and channels. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Add a batch containing three identical images in NHWC layout.

Predicted result

Prediction: Batch shape is [3, 2, 1, 3].

Why the result follows

Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Image tensors and channels to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Tensor	A numerical array with a declared set of axes
Channel	One component plane of an image such as red green or blue
Axis order	The sequence of meanings assigned to array dimensions
Value scaling	A declared numerical conversion applied to stored values

Use two terms to explain a step in this week’s example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Image tensors and channels, predict a step, then compare with its explanation.

The same pixels in three channel planes

CHANNEL	TOP ROW	BOTTOM ROW
Red c 0	[255, 0]	[0, 255]
Green c 1	[0, 255]	[0, 255]
Blue c 2	[0, 0]	[255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Use the planes to reconstruct the top-right and bottom-right RGB pixels. Give both HWC and CHW indices for the top-right green value.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Image tensors and channels

Move colour values between **tensor** layouts without changing the pixels.

The facts we will use

Supplied facts

Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/20

Two RGB pixels before layout conversion

	red	green	blue
top pixel	1	4	7
bottom pixel	2	5	8

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/20

Image tensors and channels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Record HWC shape

Shape is [2, 1, 3]; top green is 4 and bottom green5.

2. Reorder to CHW

Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].

3. Check same pixel

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

CHANGE ONE THING

Add a batch containing three identical images in NHWC layout.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record HWC shape	Shape is [2, 1, 3]; top green is 4 and bottom green5.
2. Reorder to CHW	Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].
3. Check same pixel	Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **tensor** is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself.

Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour **channel**.

What this example does not establish

Keep the stated limit

Shape consistency does not prove correct colour order or value range; document **channel**

names and pre**process**ing.

Keep the stated limit

Read the labelled picture: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Add a batch containing three identical images in NHWC layout. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Image tensors and channels claim and explain a repair using the supplied evidence.

Check the claim against the facts

For this lesson, divide every **channel** value by 255 to obtain floating-point values from 0 to 1. Thus [255, 128, 0] becomes [1, 128/255, 0], with the middle value approximately 0.502.

This fixed conversion does not fit a statistic from data. Apply the same conversion during **training** and later use, and record whether an **input** is already scaled.

Check the claim against the facts

Dividing twice makes a raw value of 255 become 1/255, approximately 0.00392, instead of 1. Integer rounding after scaling can also erase distinctions.

In exact arithmetic the declared division is reversible by multiplication; finite precision or deliberate rounding can limit exact recovery.

Check the claim against the facts

Averaging RGB **channel**s is a different operation. The classroom mean gives red [255, 0, 0] and green [0, 255, 0] the same value 85, losing their colour distinction.

It is a simple illustration, not a perceptual brightness formula. If pre**process**ing instead estimates means or spreads from a dataset, fit those statistics using **training** data only.

Locate the earliest unsupported step

Scale [255, 128, 0] by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

State one repair and the checks you would rerun.

Practise together

> Practise Image tensors and channels with support, then explain the deciding step in your own words.

For the lesson image, give image[1][0], image[1][0][2], the HWC shape, pixel count and number of **channel** values.

> **Use the planes to reconstruct the top-right and bottom-right RGB pixels. Give both HWC and CHW indices for the top-right green value.**

> **Explain your prediction to a partner before comparing results.**

Practise with less help: Audit an axis conversion

Use workbook W2 for the connected example “Image **tensor**s and **channel**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Image **tensor**s and **channel**s. Proposed explanation: “Three **channel**s mean three images”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Image tensors and channels. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

```
hwc = [
    [[255, 0, 0], [0, 255, 0]],
    [[0, 0, 255], [255, 255, 255]],
]
height, width = len(hwc), len(hwc[0])
chw = [[[hwc[r][k][c] for k in range(width)]
        for r in range(height)] for c in range(3)]
for channel in chw:
    print(channel)
assert chw[2][1][0] == hwc[1][0][2] == 255
restored = [[[chw[c][r][k] for c in range(3)]
             for k in range(width)] for r in range(height)]
assert restored == hwc
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Image tensors and channels. Explain why each step is justified.

Explain → apply → challenge

For the lesson image, give image[1][0], image[1][0][2], the HWC shape, pixel count and number of **channel** values.

> Scale [255, 128, 0] by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled image **tensor**s and **channel**s **evidence** sheet to Image-pattern Network Investigation. Show how you can read a pixel using declared row column and **channel** axes.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Stacking five same-size HWC images adds a leading example axis N. This batch has shape 5 by 2 by 2 by 3 and contains 60 **channel** values. With **channel**-first images, its shape is 5 by 3 by 2 by 2.

The example axis is not another colour **channel**. Every axis must keep its meaning through the model.

A loader returns the expected shape, but red and blue channels were exchanged and already scaled values were divided by 255 again. Does the shape check establish correctness? Name an exact pixel check and a value check that expose these faults.

Use a tiny known image, named axis meanings and expected **channel** values to check a pipeline. Keep original images when testing transformations.

Flattening can be reversed if shape and traversal order are retained; averaging colours generally cannot. Next week reads small neighbourhoods with a convolution operation.

The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Image tensors and channels case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

The same pixels in three channel planes

CHANNEL	TOP ROW	BOTTOM ROW
Red c 0	[255, 0]	[0, 255]
Green c 1	[0, 255]	[0, 255]
Blue c 2	[0, 0]	[255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Convert a new image independently

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

A batch contains four copies: give its NHWC shape and total **channel**-value count.

> **My independent explanation, trace and limits**
>
> ---
>
> ---
>
> ---
>
> ---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

A batch contains four copies: give its NHWC shape and total **channel**-value count.

> **My independent procedure, result and limitation**
>
> ---
>
> ---
>
> ---
>
> ---

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Image tensors and channels, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Move colour values between **tensor** layouts without changing the pixels.

By the end, I can	How I will show it
Read a pixel using declared row column and channel axes	Explain the deciding step, show a trace or test, and state one limit.
Convert channel layout without changing pixel meaning	Explain the deciding step, show a trace or test, and state one limit.
Distinguish fixed value scaling from information losing transformations	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Image tensors and channels without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Read a pixel using declared row column and **channel** axes

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Move colour values between **tensor** layouts without changing the pixels.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

1. Record HWC shape

2. Reorder to CHW

3. Check same pixel

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check same pixel Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. A **tensor** is an array with named dimensions.

HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself. Trace a named pixel through both layouts using zero-based indices.

Adding a batch dimension counts images and does not add a new colour **channel**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add a batch containing three identical images in NHWC layout. Which conclusion is supported by the supplied facts?

1. Batch shape is [3, 2, 1, 3].

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour **channel**.

Repair a missing building block

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

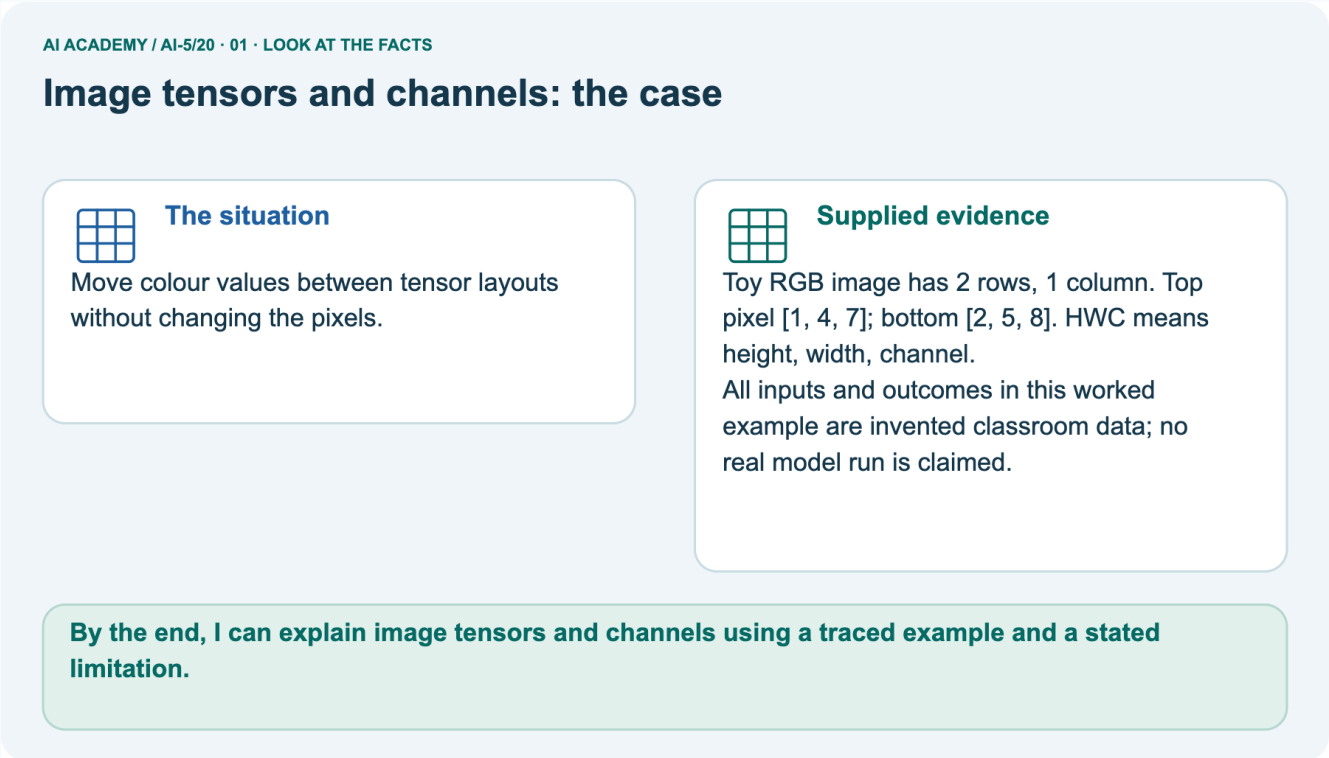
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Image tensors and channels graphic using its labels, deciding evidence and relevant limitation.

01. Image tensors and channels: the case



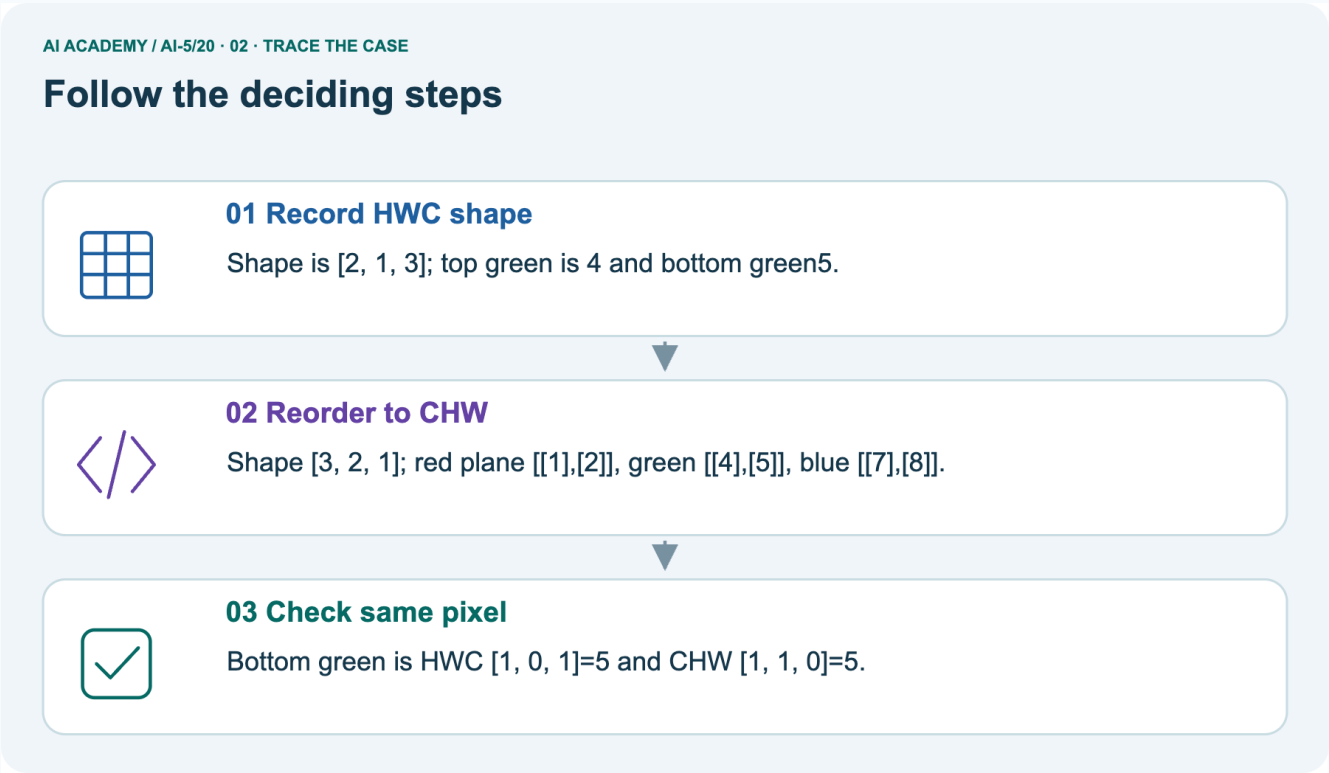
Follow the labelled picture with the matching explanation. Open the vector diagram

Move colour values between **tensor** layouts without changing the pixels. Supplied facts: Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

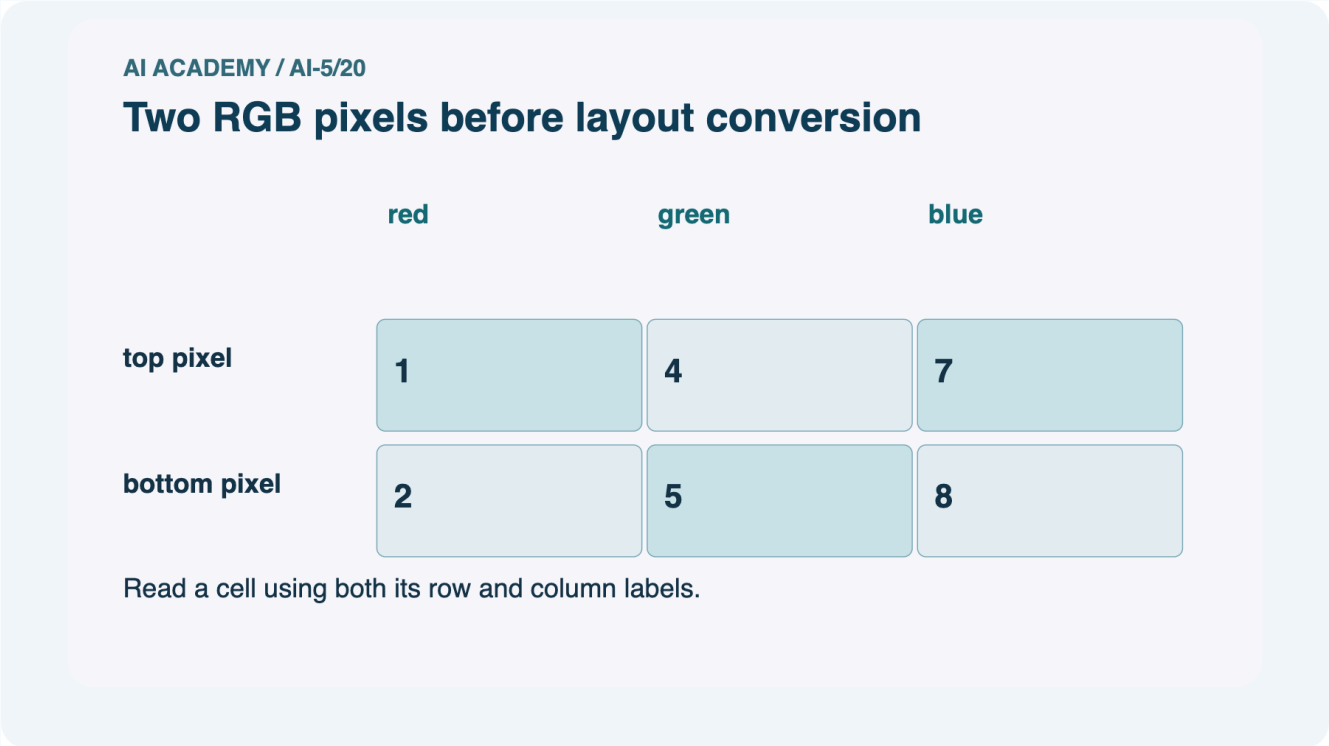


Follow the labelled picture with the matching explanation. Open the vector diagram

Record HWC shape: Shape is [2, 1, 3]; top green is 4 and bottom green5. → Reorder to CHW: Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]]. → Check same pixel: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Two RGB pixels before layout conversion

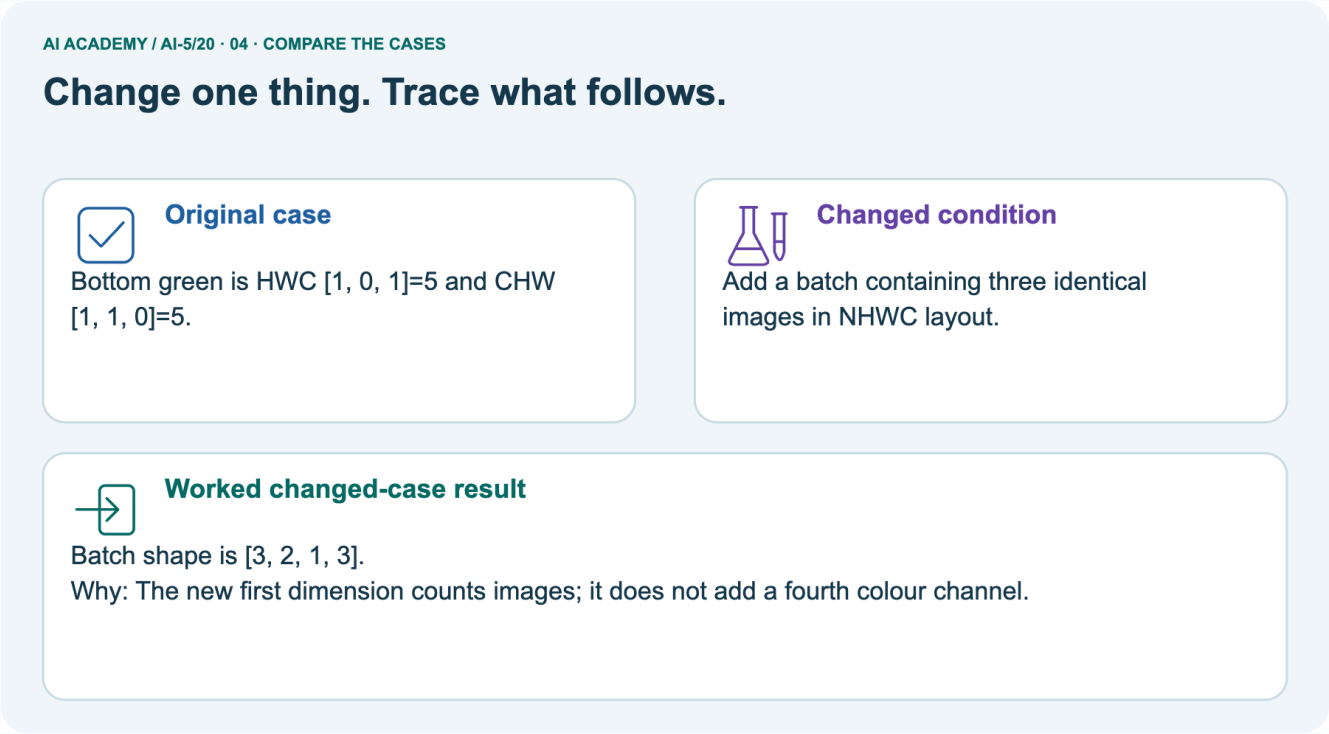


Follow the labelled picture with the matching explanation. Open the vector diagram

Two RGB pixels before layout conversion. top pixel: red = 1, green = 4, blue = 7. bottom pixel: red = 2, green = 5, blue = 8.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



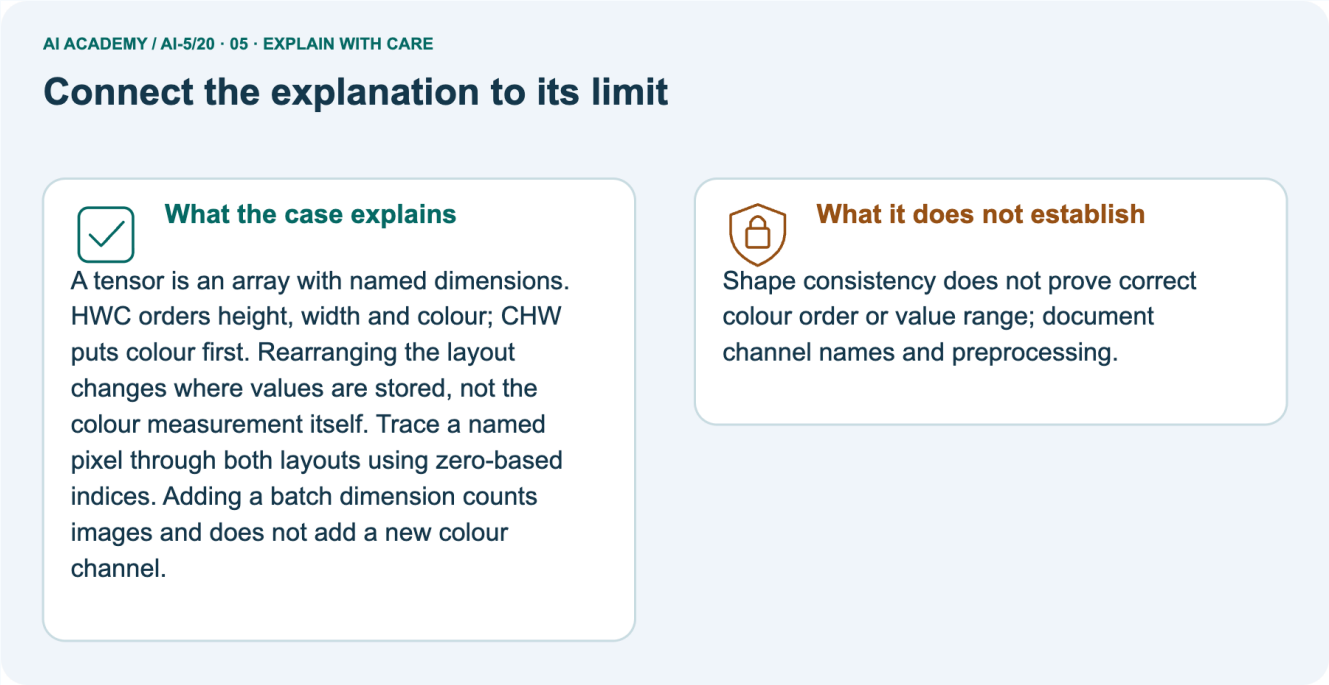
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. Changed condition: Add a batch containing three identical images in NHWC layout. Expected result: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

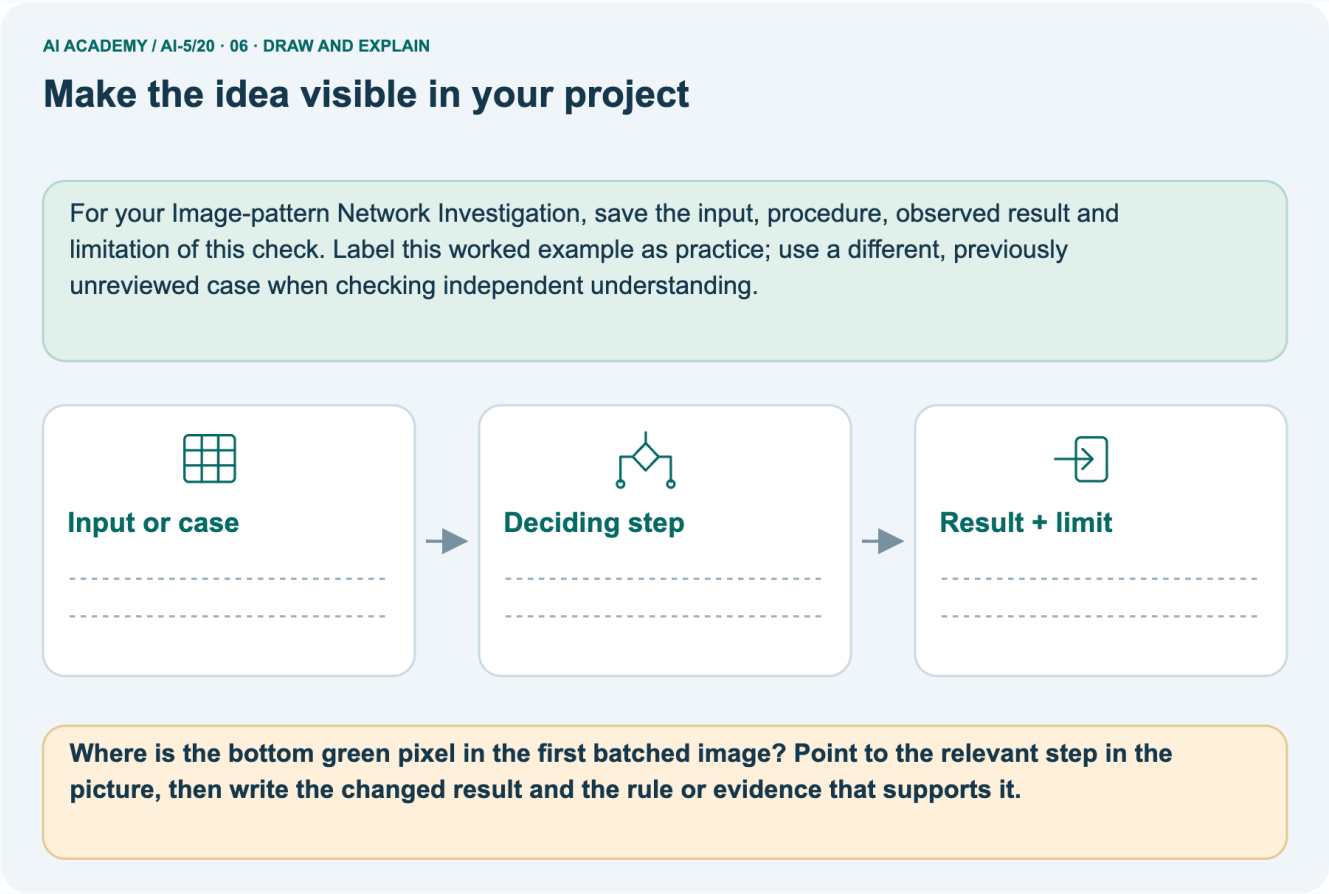
A **tensor** is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself.

Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour **channel**.

Limit: Shape consistency does not prove correct colour order or value range; document **channel** names and pre**process**ing.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.